Eric Julianto

AI Systems Engineer | Applied AI, GenAI, RAG, Computer Vision, and Cloud-Native Delivery

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Professional Summary

AI Systems Engineer building at the intersection of machine learning, generative AI, and production-grade infrastructure. Background spans LLM applications, RAG pipeline design, vision language models, forecasting research, and cloud-native deployment, with infrastructure knowledge across Dockerized services, observability stacks (Prometheus, Grafana, Loki, Tempo), messaging layers, and distributed systems. Positioned for roles requiring both model understanding and systems execution — AI Engineer, GenAI Engineer, ML Engineer, and AI Systems Engineer tracks.

Core Skills

Programming Languages: Python, JavaScript, TypeScript, Rust, Go, PHP, Elixir, Dart

AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, Time Series Forecasting, Applied Research, Model Evaluation

Generative AI: LLM, VLM, Multi-modal, RAG, Multi-Agent Systems, MCP, A2A, Prompt Engineering, LangChain, Ollama

Frameworks: FastAPI, Flask, PyTorch, React, Laravel, Actix, Phoenix

Cloud & DevOps: Docker, Kubernetes, AWS, GCP, Cloudflare, PM2, GitHub Actions, GitLab CI

Observability: Prometheus, Grafana, Loki, Tempo, Telemetry, Log Aggregation, Monitoring

Databases: PostgreSQL, MySQL, MariaDB, Redis, MinIO

Infrastructure & Systems: Linux, Ubuntu, Red Hat, RabbitMQ, VPN, Virtualized Environments

Experience

Bangkit Academy — Feb 2023 – Jan 2025

Machine Learning Mentor & Advisor

- Guided machine learning learners through model-building workflows, assignments, and applied project execution in a structured cohort environment.
- Tracked learner progress, identified technical blockers early, and delivered targeted support across core ML concepts and implementation details.
- Collaborated with instructors to refine learning materials and improve assignment clarity for a more execution-focused learning path.

Dibilabs by Dibimbing.id — Mar 2023 – Jun 2023

SEO Intern

- Performed keyword and competitor analysis to identify content opportunities and improve search visibility across target topics.
- Optimized on-page content structure, metadata, and media annotations using analytics-backed SEO workflows.
- Monitored traffic, rankings, and keyword performance using web analytics and SEO tooling to support iterative content improvements.

Indonesia AI Dec 2021 – Oct 2022

Mentor

- Delivered mentorship across data science, machine learning, deep learning, and computer vision topics in a structured community setting.
- Tracked learner progress and reported development updates to program organizers.
- Acted as a technical mentor and learning guide for students progressing through applied AI material.

ZettaByte Pte Ltd. May 2022 – Aug 2022

AI Developer

- Built AI application components including a student focus-level measurement system, question generation workflow, and question answering system.
- Deployed chatbot functionality with database integration as part of a broader AI product implementation.
- Monitored development execution, tracked progress, and followed through on delivery plans during implementation cycles.

Publications & Research

- **Improving Support Vector Machine Performance with Binary Gaussian Improved Whale Optimization Algorithm: A Case Study on Diabetes Data** — *BAREKENG: Jurnal Ilmu Matematika dan Terapan* (SINTA 2, Scopus Q4).
- **Optimizing Heart Attack Diagnosis Using Random Forest with Bat Algorithm and Greedy Crossover Technique** — *BAREKENG: Jurnal Ilmu Matematika dan Terapan* (SINTA 2, Scopus Q4).
- **Adaptive Multi-Criteria Re-Ranking for Hybrid Semantic Information Retrieval in Metadata-Rich Collections** — Scopus-indexed IEEE conference publication.
- **Automatic Plate Number Recognition (APNR) Using Deep Learning with YOLOv8 and TrOCR** — Scopus-indexed IEEE conference publication.

Selected Projects

Omniflow — AI-Native Modular Enterprise Platform Flagship project

- Designed an AI-native modular enterprise platform combining intelligent workflow orchestration, LLM-powered automation, RAG pipelines, and MCP-based agentic orchestration for enterprise process automation.
- Built distributed infrastructure spanning Dockerized services, PostgreSQL, Redis, RabbitMQ, MinIO, and an observability stack (Prometheus, Grafana, Loki, Tempo) with telemetry-driven monitoring and cloud-native deployment on Cloudflare.
- Integrated systems-thinking across modular ERP modules, AI-native automation pipelines, observability tooling, and production-oriented platform architecture with end-to-end service interoperability.
- Explored agentic system experimentation and MCP-based orchestration patterns for enterprise workflow automation, enabling multi-step task coordination and intelligent decision routing.

Tech stack: Python, JavaScript, Docker, Linux, PostgreSQL, Redis, RabbitMQ, MinIO, Grafana, Loki, Tempo, LangChain, RAG, MCP, Cloudflare, PM2

KTP OCR with VLM Selected project

- Built an OCR pipeline for extracting Indonesian ID card (KTP) data using a Vision Language Model, producing structured JSON output for e-KYC workflows.
- Engineered handling for noisy text, multi-field column layouts, and varied image quality to support downstream identity verification pipelines.
- Designed the extraction pipeline with inference-aware processing and structured output validation for production-oriented document intelligence.

Tech stack: Vision Language Model, Python, Structured Extraction, Document Intelligence, e-KYC

Finbot Selected project

- Developed a financial analysis web application for ˆJKSE and S&P 500 with real-time data retrieval, time series forecasting, and API-driven market data integration.
- Integrated news aggregation, YouTube insight extraction, and LLM-powered interactive Q&A for comprehensive financial analysis and decision support.
- Combined forecasting models with real-time data pipelines to deliver actionable market intelligence through a unified interface.

Tech stack: Python, Time Series Forecasting, Financial APIs, LLM, RAG, Web Application

Document Translator Selected project

- Built a document translation pipeline with LLM-powered translation, supporting multi-format output downloads as PDF or TXT packaged in ZIP.
- Implemented document-grounded chat functionality using retrieval-based context for follow-up questions on translated content.
- Designed the workflow around document ingestion, chunking, translation orchestration, and output packaging for scalable document processing.

Tech stack: LLM, RAG, Document Processing, PDF Generation, Translation Pipeline

Marketplace Product Recommendation System Selected project

- Built a multi-modal recommender system for marketplace products combining keyword-based text retrieval and image similarity search.
- Integrated visual and textual feature extraction to improve product discovery, relevance ranking, and cross-modal recommendation accuracy.

Tech stack: Python, Machine Learning, Computer Vision, Recommendation Systems, Vector Retrieval

Palm Oil Ripeness Detection Selected project

- Developed a specialized web application for assessing palm oil fruit ripeness with real-time image analysis and ML-based classification.
- Delivered harvesting optimization insights and interactive tools for plantation yield tracking and operational efficiency improvement.

Tech stack: Machine Learning, Computer Vision, Real-Time Inference, Web Application

AI Interview Selected project

- Built an automated interview platform using LLM-driven language and context analysis to evaluate candidate responses.
- Designed the evaluation pipeline with structured prompt orchestration and context-aware scoring for consistent, AI-driven candidate assessment.

Tech stack: LLM, NLP, Prompt Orchestration, Web Application, Recruitment Tech

Education

Universitas Esa Unggul Mar 2025 – Present

Computer Science

- Area of interest: Artificial Intelligence, Machine Learning, Computational Finance
- GPA: 4.00

Universitas Bunda Mulia Aug 2019 – May 2024

Hospitality and Tourism

- Area of interest: Tourism Management & Development and Risk Management
- GPA: 3.67
- TOEFL equivalent score: 567
- Recognition: Mahasiswa Berprestasi

Bangkit Academy led by Google, Tokopedia, Gojek & Traveloka Jan 2022 – Jul 2022

Machine Learning

- Area of interest: Machine Learning, Artificial Intelligence, Deployment
- GPA equivalent score: 4.0

AI for Startup | Orbit Future Academy Aug 2021 – Jan 2022

Data Science

- Area of interest: Data Science, Machine Learning, Artificial Intelligence
- GPA equivalent score: 4.0
- Top 16 Finalists Habibie Techfest

Speaking

- "Shifting Career To Data Science" — Webinar Coding.id.
- "Kickstart Career in Data Science" — HIMTI UIN Jakarta.
- "NVIDIA Jetson Nano for AI" — Indonesia AI Chit-chat.
- "IoT Sederhana Menggunakan Arduino" — Webinar Series Lomba THINC AI BISA AI.
- "UI/UX Design with Figma and Other Tools" — Webinar Series Lomba THINC AI BISA AI.
- "Kenalan Yuk Sama Intel Galileo" — Webinar Series Lomba THINC AI BISA AI.
- "Artificial Intelligence 101" — Forum Ilmiah Politeknik Astra.
- "Generative AI Algorithms in Mobile Application" — Tech Talk at GDSC UMS.
- "AI Integration in IoT World" — AMIKOM Computer Club.
- "Deep Dive Into the Deep Learning" — Tech Talk at GDSC UTM.
- "AI sebagai alat bantu untuk meningkatkan kreativitas manusia" — Tech Talk at Universitas Andalas.
- "Pemanfaatan Artificial Intelligence (AI) Dalam Penulisan Karya Ilmiah" — National Seminar at UM-SIDA.
- "Text2Text Generation with Gemini" — AI Workshop at UPN Jatim.
- "GDSC Forward STT NF" — Tech Talk at STT NF.
- "Designing Intelligent Recommendation Systems with Google Vertex AI" — Build with AI: Advance Workshop at GDGoC STT Terpadu Nurul Fikri.
- "AI and Environment: Paths to Regenerative Sustainability" — Tech Talk at UPB Cikarang.
- "From Code to Production: A Hands-On MLOps Workshop" — BCC FILKOM UB.